Predicting Coronary Heart Disease from Lifestyle Surveys with Regularised Logistic Regression

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***Abstract*—We predict whether a respondent of the 2015 BRFSS telephone survey has been diagnosed with coronary heart disease (MICHD) from 321 self-reported answers. Using only NumPy, we build a feature pipeline that respects the survey's coding conventions and train an L_2-regularised logistic regression with Newton's method. Tuning the decision threshold for F1 is by far the most important design choice (F1 0.237 $\rightarrow$ 0.431); feature engineering adds +0.016 over raw features, and logistic regression beats the least-squares baselines by about 0.01 F1. Our final model reaches a 5-fold cross-validated F1 of 0.431 ± 0.006 (AUC 0.861).**

I. Introduction

The dataset contains $N = 328{,}135$ training and 109,379 test respondents described by $D = 321$ raw survey variables; only 8.8% of respondents are positive. The competition metric is the F1 score of the positive class. We compare the six methods implemented in `implementations.py` with a Newton-trained logistic regression and quantify the contribution of each preprocessing step with an ablation study.

II. Data and Preprocessing

Exploration. Variables are of three kinds: (i) categorical answers coded $1, 2, \ldots$ in which 7 means "don't know" and 9 "refused"; (ii) counts or measurements with *special codes* (e.g. in `PHYSHLTH`, 88 means "zero days" and 77/99 are non-answers); (iii) survey-administration fields (interview date, sequence number, sampling weights). 99 columns are more than 90% missing because they belong to optional state modules.

Pipeline. All statistics are fit on the training data only (and re-fit inside every CV fold). (1) We drop 31 administration/weight columns and 18 raw answers with a mixed unit encoding (e.g. `FRUIT1`, where $1xx$ = per day, $2xx$ = per week) for which the CDC provides a cleaned computed variable, as well as columns with $> 90\%$ missing values. (2) Columns with ≤ 15 distinct values (and `_STATE`) are *one-hot encoded*, with a separate level for missing values; levels seen fewer than 50 times are dropped. Codes 7/9 thus become indicators instead of being read as large numbers. (3) For the 42 continuous columns we *decode special codes* (88$\rightarrow$0; 77, 99, 900, 99900$\rightarrow$missing), apply $\log(1 + x)$ to non-negative columns with skewness > 2, *clip* to the 1st–99th percentiles, impute the median, add a *missing indicator* and standardise. The result has 814 features plus a bias.

Table I
5-fold CV results. All models use a tuned threshold unless stated.

Model / variant	F1	Acc.	AUC
Baselines (implementations.py), full features			
Least squares	$X^\top X$ singular		
Ridge, $\lambda = 10^{-4}$	$.422 \pm .006$	.872	.854
Reg. logistic, GD 300 it.	$.406 \pm .003$	.860	.848
Ridge, raw features	$.410 \pm .005$	.871	.844
Newton logistic, $\lambda = 10^{-4}$ (final)	$\mathbf{.431 \pm .006}$	.871	**.861**
Ablation of the final model			
– threshold tuning ($p > 0.5$)	$.237 \pm .005$	.915	.861
– thr. tuning, + class-balanced weights	$.378 \pm .003$	.767	.861
+ class-balanced weights	$.429 \pm .007$	.872	.861
– one-hot (raw codes as numbers)	$.426 \pm .006$	.873	.859
– special-code decoding	$.430 \pm .007$	.873	.861
– log transform and clipping	$.430 \pm .006$	.871	.861
– missing indicators	$.431 \pm .006$	.873	.861
+ administration/weight columns	$.430 \pm .007$	.873	.861
+ squared continuous features	$.431 \pm .006$	.873	.861
Raw features only (mean-imputed)	$.415 \pm .005$	.868	.851

III. Models and Evaluation

Baselines. Least squares, ridge regression (normal equations, on targets $y \in \{0, 1\}$, prediction $x^\top w > t$) and regularised logistic regression trained by gradient descent (300 steps, $\gamma = 1/L$ with L the Lipschitz constant of the gradient, estimated by power iteration).

Final model. We minimise the logistic loss plus $\lambda \|w_{1:}\|^2$ (bias unpenalised) with Newton's method: each step solves $(X^\top S X/N + 2\lambda I)\, \Delta w = \nabla$ with $S = \mathrm{diag}(p_i(1 - p_i))$. It converges in 8 steps ($\sim$1 min) where gradient descent is still far from the optimum after 300 steps, which makes cross-validated model selection affordable.

Threshold. With 8.8% positives, the Bayes threshold $p = 0.5$ maximises accuracy, not F1. We choose the threshold t that maximises F1 on the training folds by sorting the scores (exact search) and apply it to the held-out fold. On the full data t corresponds to $p = 0.20$.

Protocol. Stratified 5-fold CV (seed 0) for every number below; we report the mean $\pm$ std over folds of F1, accuracy and ROC-AUC.

IV. Results

Table I and Fig. 1 summarise the experiments.

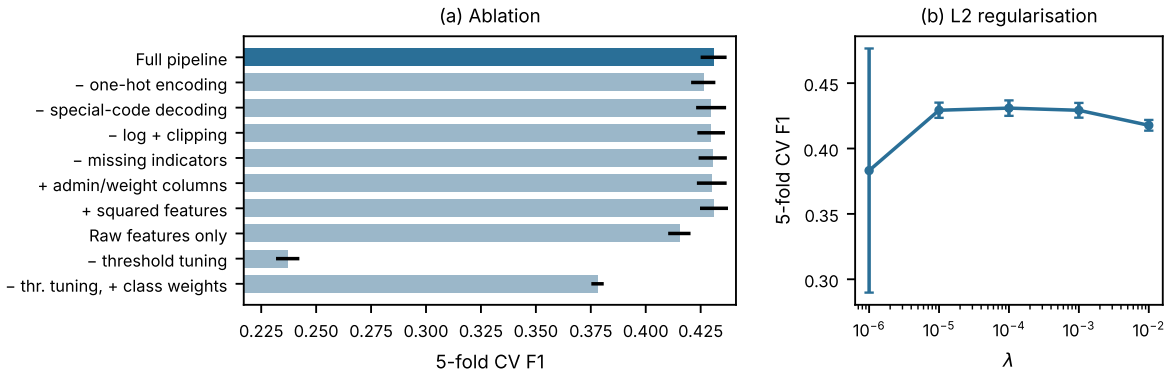


Figure 1. (a) 5-fold CV F1 of the ablation variants (bars: std over folds). (b) F1 versus the L_2 penalty λ of the Newton logistic regression.

Threshold is the dominant factor. Predicting $p > 0.5$ yields the best accuracy (0.915) but an F1 of only 0.237, because few respondents reach $p > 0.5$. Re-weighting the classes recovers part of the gap (0.378), but tuning the threshold directly gives 0.431; once the threshold is tuned, class weights bring nothing (0.429): the AUC is unchanged, so the ranking of respondents is the same and only the operating point moves.

Model. Logistic regression outperforms ridge regression by +0.009 F1 and +0.007 AUC on identical features: the squared loss penalises confident correct predictions and is ill-suited to a rare binary target. Plain least squares fails outright: every one-hot block (including its missing level) sums to the bias column, so $X^\top X$ is singular. The same collinearity explains Fig. 1(b): for $\lambda = 10^{-6}$ the Hessian is nearly singular and the undamped Newton iteration becomes unstable on some folds (F1 0.38 ± 0.09), whereas $\lambda \in [10^{-5}, 10^{-3}]$ gives a flat optimum; $\lambda = 10^{-2}$ underfits (0.418). Gradient descent with the same objective reaches only 0.406 after 300 steps, showing that the optimiser, not the model, limits the GD baseline.

Feature engineering. The full pipeline improves F1 by +0.016 and AUC by +0.010 over raw mean-imputed features, for both ridge and logistic regression. One-hot encoding is the most useful single step (-0.005 when removed). Removing any other single step costs at most 0.001, below the fold-to-fold std: the steps are partly redundant (e.g. decoding is only needed for 13 continuous columns since one-hot already isolates codes 7/9 in categorical ones). Squared features bring $+0.0002$ for 42 extra columns and were not kept. Keeping the administration columns does not help either, which is reassuring: the model does not exploit artefacts of the survey process.

Interpretation. The largest coefficients correspond to a previous stroke, male sex, poor self-rated general health, COPD, kidney disease, age and estimated cardiorespiratory fitness (`MAXVO2_`, negative) — all known cardiovascular risk factors.

V. Conclusion

A carefully regularised linear model, trained with a second-order optimiser and evaluated with the metric's own decision rule, reaches a CV F1 of 0.431. Choosing the operating point mattered far more than any single preprocessing step; the remaining gap is likely due to the linear decision boundary, which non-linear models (Project 2) could address. Our submission is reproduced by `run.py`; every number above is produced by `experiments.py`.